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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

November 2022 - Paper 2 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-FS-Maths-L2-Paper-2-November-2022

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Section A

Answer **all** questions in the spaces provided.

1 Circle the integer.

[1 mark]

0.5 $\frac{1}{8}$ 7 −10.2

Q1 Worked Solution:

Nine million, five hundred (and) seven thousand, two hundred (and) eleven

2 Write 9 507 211 in words.

[1 mark]

Answer _____

Q2 Worked Solution:

Ignore spelling if intention is clear

Ignore punctuation and grammar

4 : 1

1 :

1 : 0.25

[B1] [B2]

3 Work out 3 years to 9 months as a ratio.

Give your answer in its simplest form.

[2 marks]

Q3 Worked Solution:

Method and answer verified against official AQA mark scheme.

[Method & Accuracy]

Answer _____ : _____

Question 1 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

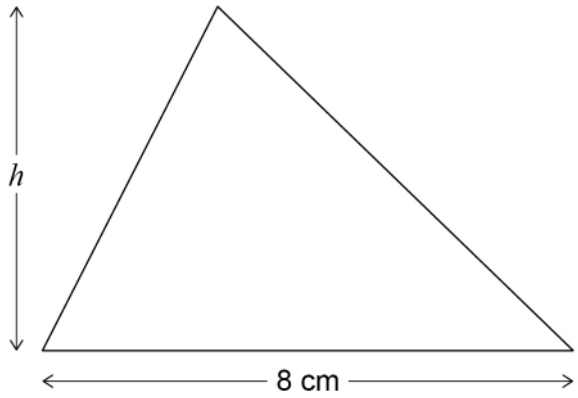
[B1] [B1]

Do not write outside the box

7 A triangle has an area of 20 cm^2

The base of the triangle is 8 cm

Not drawn accurately



Q7 Worked Solution:

20 × 2 or 40
20 8 or 2.5
Alternative method 1
−1.8 −5.4
3 × or
+ × −1.8
7 3 or 1.6
3.2
[A1] [M1] [A1] [SC1]

Work out the perpendicular height, h, of the triangle.

[2 marks]

Answer _____ cm

8 Calculate $2(7 + 3k)$ when $k = -1.8$

[2 marks]

Q8 Worked Solution:

+ 6k
14
−1.8 −10.8
6 or
3.2

Answer _____

Alternative method 1 – converting km
Any correct km to mile conversion
using the graph
8 5
1.6 or 0.625 or or oe seen
5 8

12

Turn over ►

Question 8 · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Uses their conversion to work out

[M1] [A1] [SC1] [M1]

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

9 **Lorry driving**

Asha is a lorry driver.

9 (a) Asha drives from Southampton to Leeds.

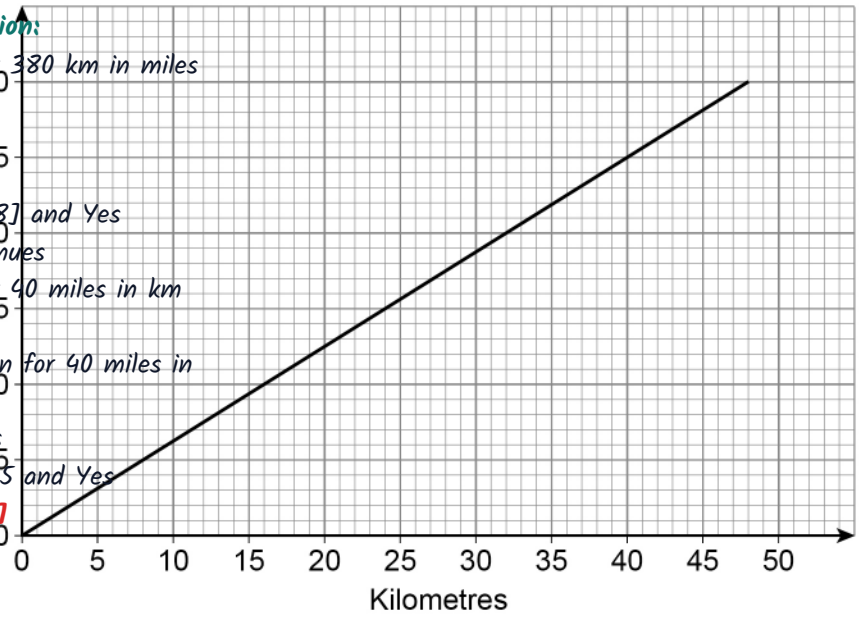
The journey

is 380 kilometres

takes 5 hours and 30 minutes.

The graph can be used to convert between miles and kilometres.

Q9(a) Worked Solution:
their conversion for 380 km in miles
5.5
40 5.5 or 220
[41, 45.1] and Yes
220 and [228, 248] and Yes
Mark scheme continues
their conversion for 40 miles in km
5.5 or [341, 374]
380 their conversion for 40 miles in km or [5.58, 6.13]
[341, 374] and Yes
[5.58, 6.13] and 5.5 and Yes
[M1] [A1] [A1] [A1]



Asha works out that his average speed is **over** 40 mph

Is he correct?

You **must** show your working.

[4 marks]

Do not write outside the box

9 (b) The amount Asha is paid each week is calculated using the formula

$$P = 0.73(0.14d + 65n)$$

where

P = pay in pounds

d = distance driven in kilometres that week

n = number of days worked that week

Last week Asha worked for 5 days.

His pay for last week's work was £605.17

How many kilometres did Asha drive last week?

[4 marks]

Answer _____ kilometres

8

Turn over for the next question

Do not write outside the box

10 Fundraising

Carol is fundraising for a sports club.

10 (a) Carol designs a game.

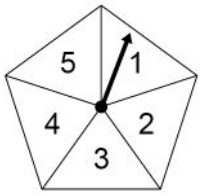
The game uses

a bag containing a red ball, a blue ball and a yellow ball

a fair, 5-sided spinner.

Q10(a) Worked Solution:
20 15
and and Yes
150 150
[A1]





The player

picks a ball at random from the bag

and

spins the spinner.

The player wins if they pick the **red** ball and the spinner lands on an **even** number.

Carol says,

"The chance of winning is **more than** 10%"

Is she correct?

You **must** show your working.

[4 marks]

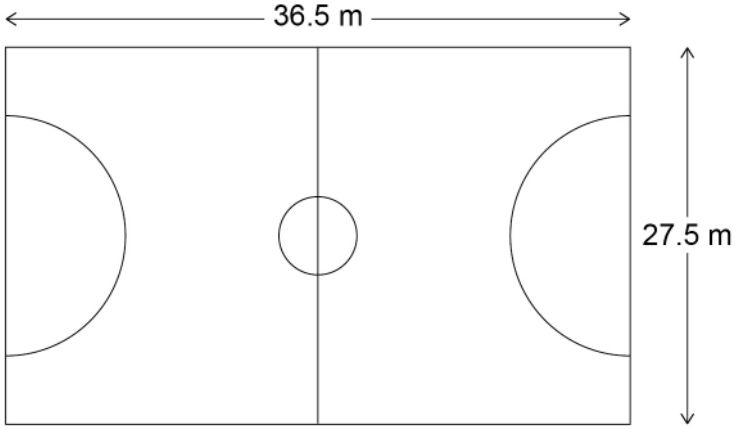
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10 (b) The club wants to use some of the money to paint the lines on a mini football pitch.

The lines to be painted are

- the four sides of a rectangle measuring 27.5 m by 36.5 m
- a halfway line measuring 27.5 m
- a centre circle with a radius of 1.5 m
- two semicircles, each with a radius of 8 m

Not drawn accurately



It costs £3.25 per metre to paint the lines.

In total, how much will it cost to paint all the lines?

[6 marks]

Answer £ _____

Turn over ►

Do not write outside the box

10 (c) After painting the lines the club has £8225

They invest $\frac{2}{7}$ of this money in a bank account for 4 years.

The account pays compound interest at 3% per year.

Is the investment worth **more than** £2700 at the end of the **4** years?

You **must** show your working.

[4 marks]

14

Do not write outside the box

11 **Ice cream**

Suzi has an ice cream van.

11 (a) Suzi buys tubs of ice cream and sells scoops of ice cream.

Each scoop is in the shape of a sphere with radius 2.8 cm

$$\text{volume of sphere} = \frac{4}{3}\pi r^3$$

r = radius of sphere

Suzi buys 5-litre tubs.

1 litre = 1000 cm^3

Suzi wants to buy enough tubs to sell at least 200 scoops.

Work out how many tubs Suzi should buy.

You **must** show your working.

[5 marks]

Q11(a) Worked Solution:

[18 000, 18 400] and

x x =

5 1000 4 20 000

[3.6, ~~3.7~~] and 4

[54, 56] and 4

x =

[18, 18.4] and 5 4 20 and 4

5 1000 4 20 000 and 4

Alternative method 1

76.5(0) 0.85 or 90

their 90 76.5(0)

their 90 14

[A1] [M1] [A1]

Answer _____

Question 11 continues on the next page

Do not write outside the box

11 (b) Suzi uses a 15% discount voucher when she buys the tubs of ice cream.

She pays £76.50 after the discount.

Suzi says,

"I save **less than** £14 by using the discount voucher."

Show working to support this statement.

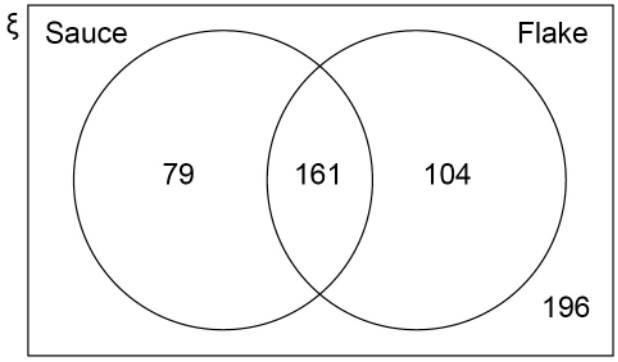
[3 marks]

Do not write outside the box

11 (c) Suzi sells two ice cream toppings, sauce and flake.

She hopes that the probability that a customer, picked at random, buys **at least one** topping will be more than $\frac{7}{10}$

The Venn diagram shows what toppings the customers buy over one weekend.



Over this weekend, does Suzi achieve the probability she hopes to get?

You **must** show your working.

[3 marks]

11

Turn over for the next question

Do not write outside the box

12 **Bees**

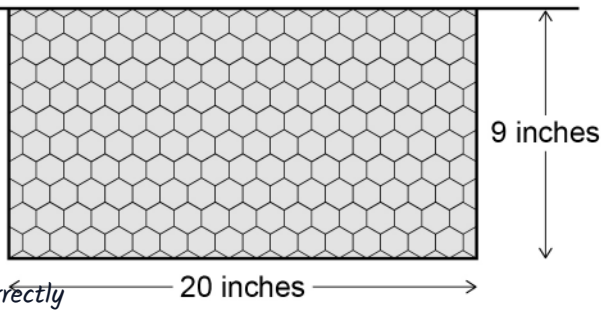
Mary keeps bees and sells the honey they produce.

12 (a) The bees live in a beehive.

Mary's beehive holds rectangular frames full of honeycomb.

Q12(a) Worked Solution:

÷ ÷
20 (7.5 2.5) or 6.66... or 6
and
9 (11 2.5) or 2.045... or 2
12
their 12 8
96
Alternative method 1
Any two angles measured correctly
[43, 47] or [58, 62] or [118, 122] or
[33, 37] or [63, 67]
+ +
[M1] [A1] [M1] [A1] [M1]



Not drawn accurately

Each frame measures 9 inches by 20 inches.

The beehive holds 8 frames.

Mary cuts the honeycomb into rectangular pieces measuring 11 cm by 7.5 cm

Work out the maximum number of pieces that Mary can get from her **beehive**.

Use 1 inch = 2.5 cm

[5 marks]

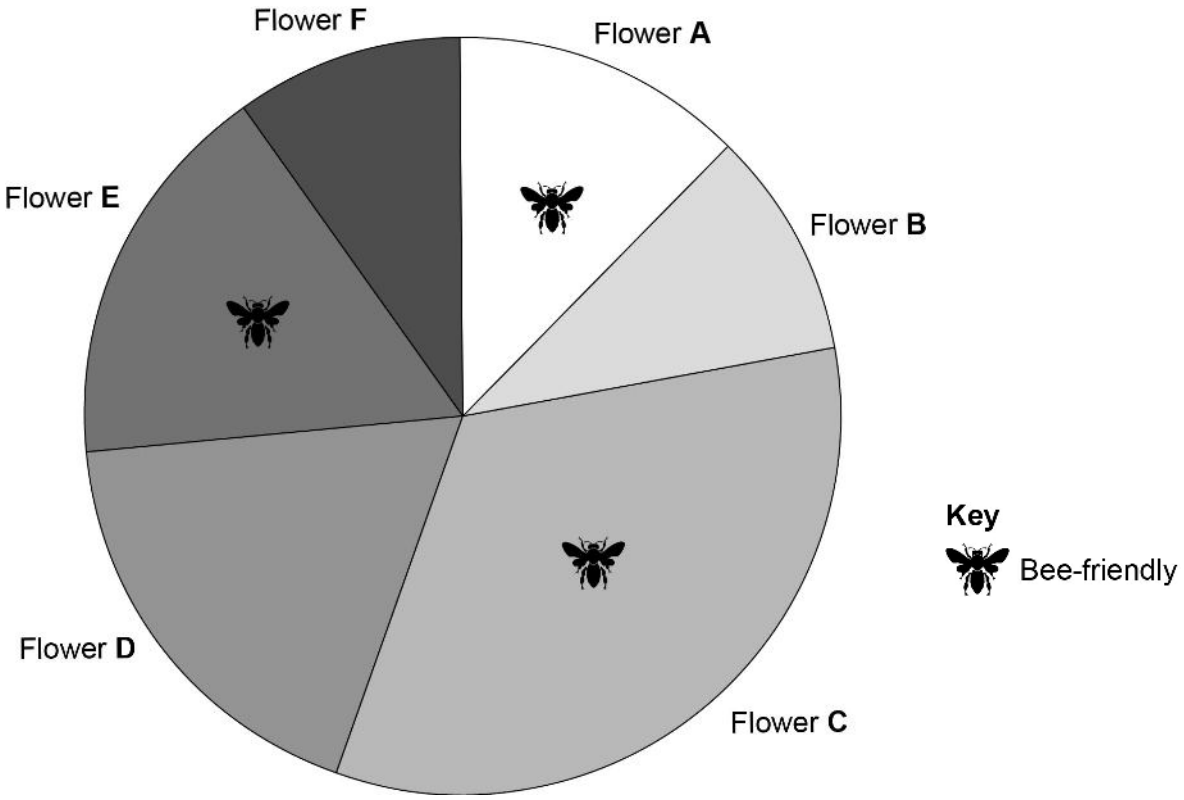
Answer _____

Do not write outside the box

12 (b) Mary wants to grow some bee-friendly flowers.

She finds information about the different flowers produced from two packets of seeds.

Packet 1



Two thirds of the seeds in **Packet 2** produce bee-friendly flowers.

Mary wants to buy the packet producing the greater proportion of bee-friendly flowers.

Which packet should she buy?

You **must** show your working.

[4 marks]

Do not write outside the box

12 (c) Here are the instructions for planting flower seeds.

Use 4 grams of seed per square metre of garden.

Mix the seeds with sand in the ratio

mass of seed : mass of sand = 2 : 5

Mary measures her neighbours' gardens to work out the average-sized garden.

Garden (m^2)	Frequency	Midpoint	
$0 < \text{area} \leqslant 10$	2		
$10 < \text{area} \leqslant 20$	8		
$20 < \text{area} \leqslant 30$	12		
$30 < \text{area} \leqslant 40$	3		

Estimate the total mass of seed and sand mix needed to cover an average-sized garden.

[6 marks]

Answer _____ grams

15

END OF QUESTIONS

Do not write outside the box

Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**

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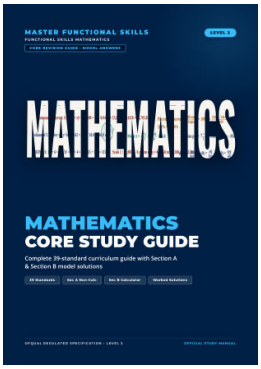
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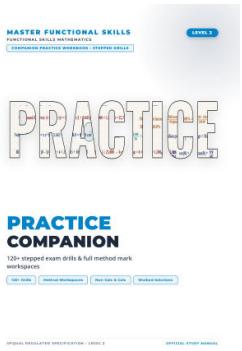


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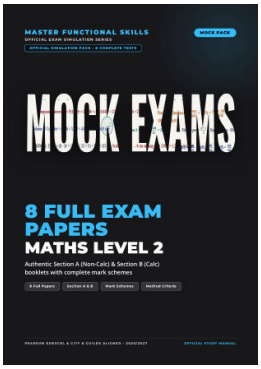


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